

DevConnect - Product Requirements Document (PRD)

Project Overview

DevConnect is a developer-focused social networking platform where users can create profiles, share posts, follow other developers, interact through likes and comments, and manage authenticated sessions across devices.

The goal of this project is to learn production-grade full-stack development using React, Node.js, Express, MongoDB, JWT Authentication, Session Management, and Cloudinary.

Tech Stack

Frontend

- React
- React Router
- Context API
- Axios
- Tailwind CSS

Backend

- Node.js
 - Express.js
 - MongoDB
 - Mongoose
 - JWT
 - Cookie Parser
 - Cloudinary
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User Roles

User

Can:

- Register

- Login
- Create Posts
- Edit Own Posts
- Delete Own Posts
- Like Posts
- Comment on Posts
- Follow Users
- Update Profile

Admin

Can:

- View All Users
 - Ban Users
 - Delete Users
-

Authentication Module

Features

Register

Fields:

- Username
- Email
- Password

Flow:

1. User submits form
2. Account created
3. OTP sent to email
4. User verifies email

Login

Fields:

- Email
- Password

Flow:

1. Verify credentials
2. Create session
3. Generate Access Token
4. Generate Refresh Token
5. Login successful

Logout

Flow:

1. Revoke current session
2. Clear refresh cookie

Logout All Devices

Flow:

1. Revoke all active sessions
2. Clear cookie

Refresh Token

Flow:

1. Refresh token received from cookie
2. Verify session
3. Generate new access token
4. Rotate refresh token

User Profile Module

Fields:

- Profile Image
- Username
- Bio
- Skills
- Location

Features:

- Update Profile
- View Profile
- Upload Profile Picture

Post Module

Create Post

Fields:

- Text
- Optional Image

Update Post

Owner only

Delete Post

Owner only

Get Feed

Returns latest posts

Pagination required

Follow Module

Features:

- Follow User
 - Unfollow User
 - Followers Count
 - Following Count
-

Like Module

Features:

- Like Post
- Unlike Post

Rules:

- One user can like a post only once
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Comment Module

Features:

- Add Comment
 - Delete Own Comment
 - Get All Comments
-

Search Module

Features:

- Search User By Username
 - Search User By Skill
-

Session Management Module

Store:

- User
- Refresh Token Hash
- Device Name
- User Agent
- IP Address
- Created At
- Last Active At

Features:

- Active Sessions List
 - Logout Specific Device
 - Logout All Devices
-

Premium Device Limitation

Free Plan:

- Maximum 1 Active Device

Premium Plan:

- Maximum 2 Active Devices

Flow:

1. User logs in
2. Count active sessions
3. If limit exceeded:
4. Revoke oldest session
5. Create new session

Admin Module

Features:

View Users

- Paginated

Ban User

- Prevent Login

Delete User

- Remove User Data

API Endpoints

Auth

POST /api/auth/register

POST /api/auth/verify-email

POST /api/auth/login

POST /api/auth/logout

POST /api/auth/logout-all

POST /api/auth/refresh-token

GET /api/auth/me

User

GET /api/users/:id

PATCH /api/users/profile

PATCH /api/users/avatar

Posts

POST /api/posts

GET /api/posts

GET /api/posts/:id

PATCH /api/posts/:id

DELETE /api/posts/:id

Follow

POST /api/follow/:userId

DELETE /api/follow/:userId

GET /api/followers/:userId

GET /api/following/:userId

Likes

POST /api/posts/:postId/like

DELETE /api/posts/:postId/like

Comments

POST /api/posts/:postId/comments

DELETE /api/comments/:commentId

GET /api/posts/:postId/comments

Admin

GET /api/admin/users

PATCH /api/admin/users/:id/ban

DELETE /api/admin/users/:id

Database Collections

Users

Sessions

Otps

Posts

Comments

Likes

Follows

Project Success Criteria

A user should be able to:

1. Register
2. Verify Email
3. Login
4. Create Profile
5. Create Post
6. Follow Users
7. Like Posts
8. Comment on Posts
9. Manage Sessions
10. Logout From Specific Devices
11. Logout From All Devices

Project should be fully deployed and accessible publicly.